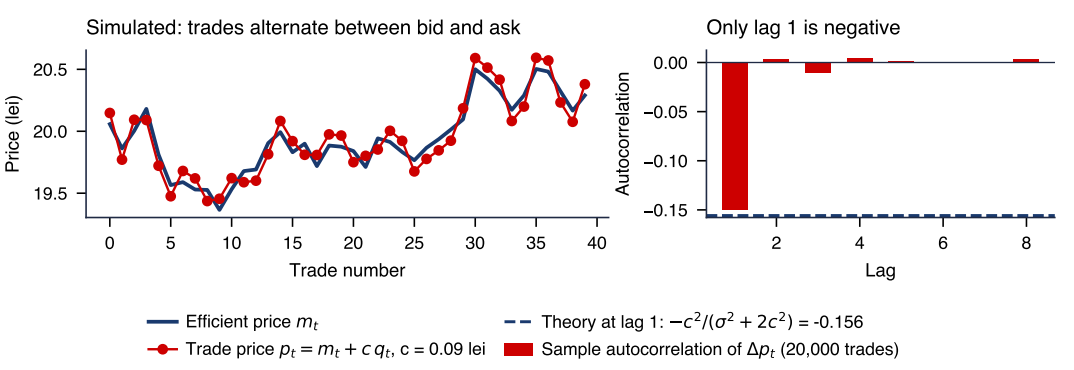
Simulated: trades alternate between bid and ask
Price (lei)
20.5
20.0
19.5
0
5
10
15
20
25
30
35
40
Trade number
Only lag 1 is negative
Autocorrelation
0.00
−0.05
−0.10
−0.15
2
4
6
8
Lag
Efficient price m_t
Trade price $p_t = m_t + c\,q_t$, c = 0.09 lei
Theory at lag 1: $-c^2/(\sigma^2 + 2c^2)$ = -0.156
Sample autocorrelation of Δp_t (20,000 trades)